L'Hôpital's Rule may be used **only** for limits of the form

$$\frac{0}{0} \text{ or } \frac{\infty}{\infty}$$

but there are many other indeterminate forms, such as :

$$0 \cdot \infty, 0^0, 1^\infty, \infty^0, \text{ or } \infty - \infty$$

Recall that these are all related to limits so that none of the pieces involved are identically 0, 1, or ∞. Rather, we consider expressions which approach 0, 1, or ∞. The value of the limit depends on the rates at which these expressions approach the values in question.

Evaluate the following limits.

1. $\lim_{x \to 0} \frac{\sin(x)}{e^x - e^{-x}}$

2. $\lim_{x \to 0} \frac{1 - \cos(3x)}{2x + 3x^2}$

3. $\lim_{x \to 0} \frac{3x + \sin(2x)}{4x}$

4. $\lim_{x \to \infty} \frac{2^x}{x^2}$

For the following problems, rewrite the expression so that it is in L'Hôpital-ready form. Then evaluate the limit.

5. $\lim_{x\to 0} x \cdot \cot(3x)$

6. $\lim_{x\to 0^+} x^2 \cdot \ln(x)$

For indeterminate forms with exponents, we need to take another approach. Consider the following example with a as a constant:

$$\lim_{x\to 0} (1+ax)^{1/x}$$

First, let $y = (1+ax)^{1/x}$. Then take the natural log of both sides and use properties of logs to simplify the right side.

$$\ln(y) =$$

Find the limit of both sides. (Use L'Hôpital's Rule).

$$\lim_{x\to 0} (\ln(y)) =$$

Then, since $\ln(y) \to$ _____, then $y \to$ _____.

7. Find the following limits. (No work required here - use a pattern)

   (a) $\lim_{x\to 0} (1+3x)^{1/x}$

   (b) $\lim_{x\to 0} \left(1+\frac{x}{4}\right)^{1/x}$

   (c) $\lim_{n\to\infty} \left(1+\frac{2}{n}\right)^{n}$

More practice. Show all work here.

8. $\lim_{x\to 0^+} x^{\sin(2x)}$

9. $\lim_{x\to 0} (e^x + x)^{1/x}$

10. $\lim_{x\to 1}\left(\dfrac{x}{x-1}-\dfrac{1}{\ln x}\right)$

11. (a) Determine conditions on p so that $\lim_{x\to\infty}\dfrac{\ln x}{x^p}=0$

(b) Evaluate $\lim_{x\to\infty}\left(x^2-\ln(x)\right)$ **Hint:** Factoring may be helpful.

(c) Based on your previous answers, what can you say about the rate at which x^p (for $p>0$) and $\ln(x)$ approach infinity?